Unit no. ° 5

TERM REVIEW AND REMEDIATION

16-20 November 2026

I Individual review

Instructions

Work without the notes for 25 minutes, then check with the methods of the booklet. For each question, record: succeeded alone, succeeded with help, or to be revisited. Justify the answers.

Example 5.1: Review of units 1 to 4

1. Make 90 /126 in lowest terms. Prove that the sum of two odd integers is even.

2. With $A(-1;2)$ and $B(3;-2)$, calculate $\overrightarrow{AB}$, AB and the midpoint of $[AB]$ in an orthonormal coordinate system.

3. Determine an equation of (AB) ; $C(5;-4)$: does it lie on this line?

4. Calculate $[-2;4]\cap]1;7]$ and $[-2;4]\cup]1;7]$.

5. Solve $|x-1|\leq 4$ and explain why 1 /3 $\notin \mathbb{D}$.

Solution:

1. GCD $(90, 126) = 18$, so 90 $/126 = 5/7$. If $u = 2k + 1$ and $v = 2\ell + 1$, then $u + v = 2(k + \ell + 1)$ is even.
2. $AB = (4;-4)$; $AB = 4\sqrt{2}$; midpoint $I(1;0)$.
3. $y = -x + 1$. For C, $-4 = -5 + 1$: it lies on (AB).
4. Intersection $]1;4]$; union $[-2;7]$.
5. $x \in [-3;5]$. Writing 1 $/3 = a/10^n$ would require that 10^n is divisible by 3, which is impossible.

Guidance for the teacher

Possible marking out of 20: 4 points per question. Record separately mastery of calculation, choice of method and quality of justification.

II Correct a mistake and redo a method

Example 5.2: Analysing pupils' work

Correct: (a) "$\overrightarrow{AB} = (x_A - x_B; y_A - y_B)$"; (b) "The zero vector directs every line"; (c) "$|x-2| \leq 3$ gives $x \in [2;3]$" ; (d) "$1/3 \approx 0{,}333$, hence $1/3$ is a decimal".

Solution:

(a) Subtract the initial point from the end point: ($x_B - x_A; y_B - y_A$). (b) A direction vector must be non-zero. (c) The distance from 2 is at most 3: $x \in [-1;5]$. (d) $0{,}333$ is different from $1/3$: $3 \times 0{,}333 = 0{,}999 \neq 1$. A decimal approximation does not determine the nature of the number being approximated.

Example 5.3: Review pathway

1. Simplify $72/108$, then $\frac{3}{4} - \frac{1}{6}$.

2. With $P(1;1)$ and $Q(5;3)$, find the midpoint, a direction vector and an equation of (PQ).

3. Determine $]-\infty;3] \cap [-1;+\infty[$, then solve $|x+2| < 1$.

Solution:

1. $2/3$; $7/12$. 2. Midpoint (3; 2), direction vector (2; 1), equation $x - 2y + 1 = 0$. 3. Intersection $[-1;3]$; solution of the inequality $]-3;-1[$.

III Applying and self-assessing

Example 5.4: Synthesis problem

Two displacements are represented by $\vec{u}(6; 4)$ and $\vec{v}(9; 6)$.

1. Show that they are collinear and find the coefficient linking $\vec{v}$ to $\vec{u}$.

2. The start is $A(-2; 1)$. Determine the end points B and C associated with the displacements $\vec{u}$ and $\vec{v}$.

3. Find an equation of the line through these three points.

4. A point $M(x; y)$ of this line must satisfy $|x - 4| \leq 3$. Give the interval of possible x-coordinates.

Solution:

1. $\vec{v} = \frac{3}{2}\vec{u}$. 2. $B(4; 5)$ and $C(7; 7)$. 3. A direction vector is (3; 2); the equation is $2x - 3y + 7 = 0$, satisfied by the three points. 4. $|x - 4| \leq 3 \iff x \in [1; 7]$.

My review

My priority: . . . . . .
Mistake to avoid: . . . . . .
Method to reuse: . . . . . .
An exercise to redo without help: . . . . . .

Guidance for the teacher

Confirm the review with a similar exercise done without help after a delay. Orally, ask for a sentence linking the calculation and the geometric conclusion.